

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Thomas Kwa, Ben West, Joel Becker et al. — METR (Model Evaluation & Threat Research)
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~7 mo

Doubling Time

$R^2 = 0.97$

Trend Fit

2 sec → 2 hrs

GPT-2 → o3

A New Metric: 50% Task-Completion Time Horizon

The duration of tasks (measured by skilled human time)
that an AI model can complete with 50% success rate.

Three-step methodology:

1 Assemble Tasks

Curate 170 diverse
software & research tasks
(3 sec – 8 hrs human time)

2 Measure

Time skilled humans &
measure AI success rates
(800+ baselines, 2,529 hrs)

3 Fit Model

Fit logistic curve to find
task duration where AI
succeeds 50% of the time

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST Diverse software tasks 1 min – 30 hrs 97 tasks	RE-Bench ML research engineering 8 hrs (all) 7 tasks	SWAA Single-step atomic actions 1 sec – 30 sec 66 tasks
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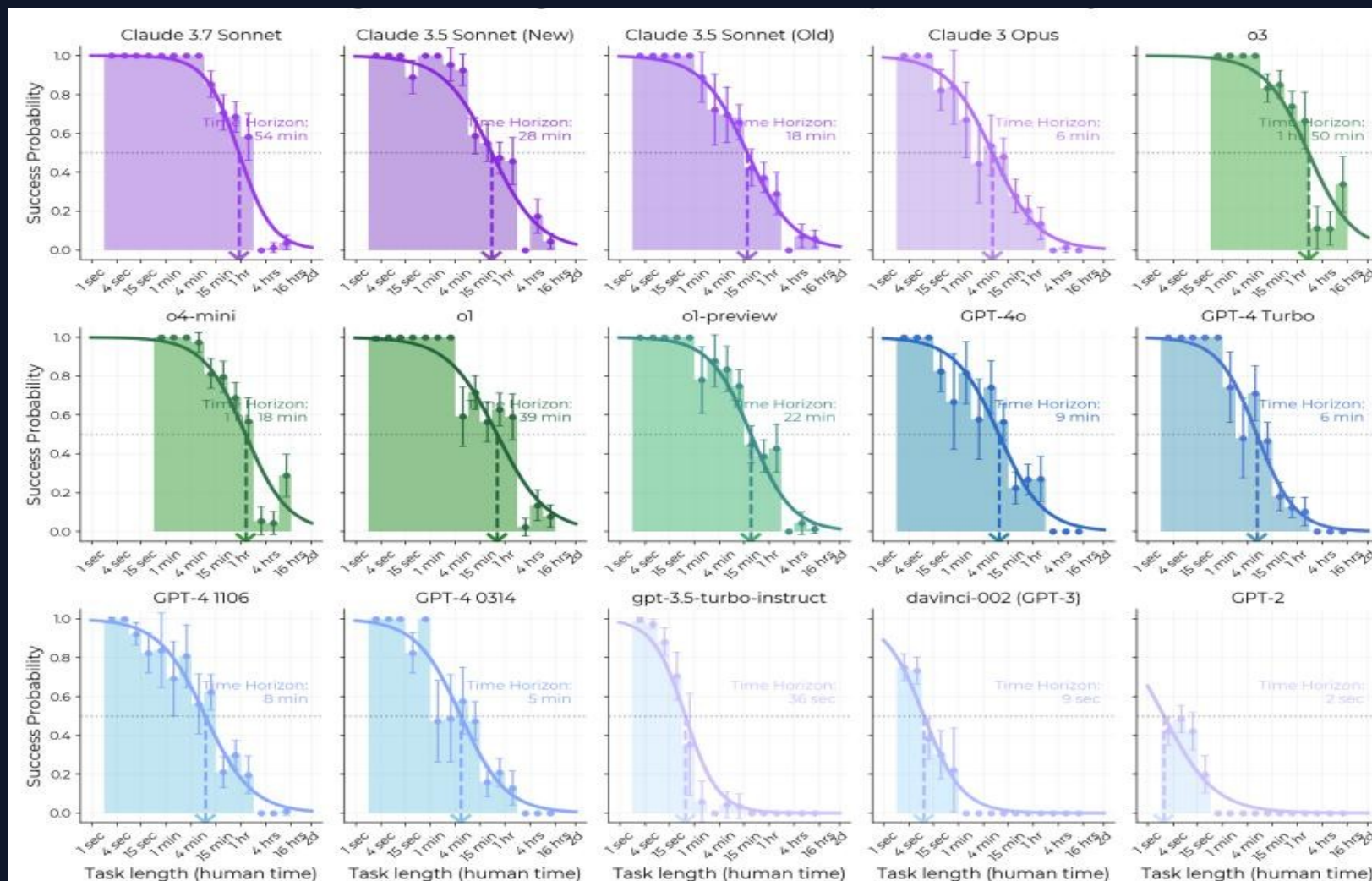
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oxdna_simple	9 minutes	Detect & fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of backtesting tool

AI Task Horizon: 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years

Doubling Time: ~207 days (~7 months)

$R^2: 0.97$

2023-2025 rate: ~20% faster than overall trend



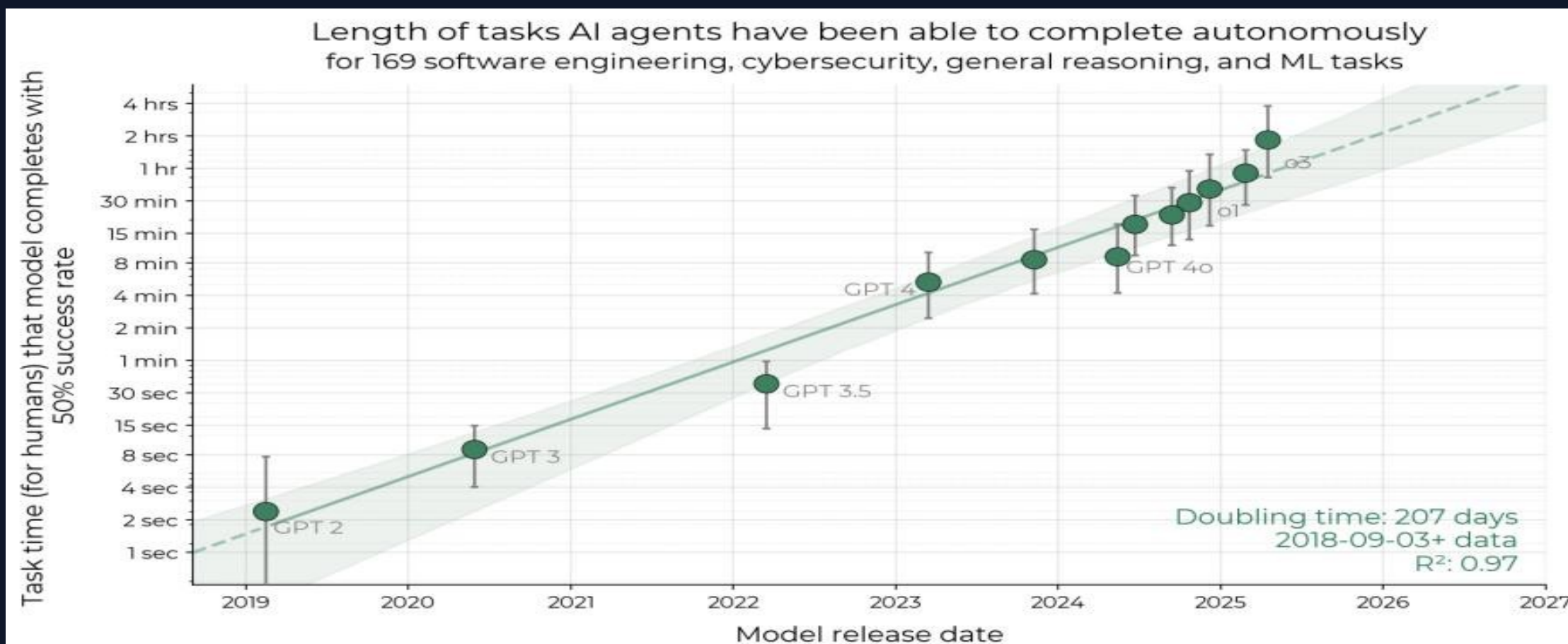
Model Milestones

GPT-2 (2019)	~2 sec
GPT-3 (2020)	~9 sec
GPT-3.5 (2023)	~36 sec
GPT-4 (2023)	~5 min
GPT-4o (2024)	~9 min
Claude 3 Opus (2024)	~6 min
Claude 3.5 Sonnet (2024)	~28 min
o1 (2024)	~39 min
Claude 3.7 Sonnet (2025)	~54 min
o3 (2025)	~1 hr 50 min

Success Probability Drops Sharply as Task Length Increases

Each curve shows a model's success rate vs. human task duration. Frontier models push the curve rightward.

The 80% time horizon follows a similar trend but with horizons roughly 5× shorter.



Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Improved Logical Reasoning

Models demonstrate stronger multi-step deduction and planning, enabling them to decompose complex tasks into manageable sub-problems.

Greater Reliability & Self-Awareness

Modern agents recognize when they are stuck, re-evaluate their approach, and self-correct rather than persisting with failing strategies.

Better Tool Use

Frontier models more effectively leverage shells, APIs, file systems, and external tools to execute real-world software engineering workflows.

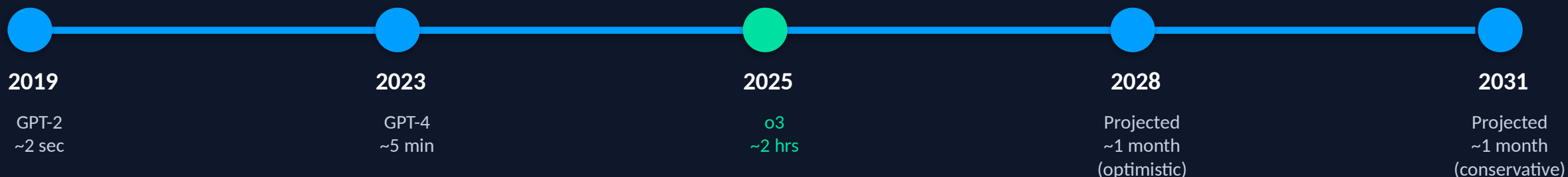
Ability to Adapt to Mistakes

Error recovery capability lets agents debug their own outputs, interpret error messages, and iteratively refine solutions.

Key Limitation: Performance is notably lower on less structured, “messier” tasks. Supplementary experiments found little evidence that trends are slower on more realistic tasks, but external validity remains an open question.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

If the exponential trend continues, AI agents will reach a 1-month time horizon (167 work hours) between mid-2028 and mid-2031.



- 2023–2025 growth rate is ~20% faster than the overall 2019–2025 rate, suggesting possible acceleration
- 167 work hours ≈ one month of full-time skilled labor
- Extrapolation is naïve: assumes trend continuity — real progress may plateau or accelerate
- External validity caveat: benchmark tasks may not represent average real-world intellectual labor

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust Metric

50% time horizon provides an intuitive, human-grounded metric for tracking AI autonomous capability with $R^2 = 0.97$ over 6 years of model releases.

2

Rapid Doubling

The time horizon doubles every ~7 months, with possible acceleration since 2024. This pace outstrips most governance and safety evaluation cycles.

3

Current Frontier

Frontier agents (o3) autonomously complete tasks requiring ~2 hours of skilled human effort, up from 2 seconds (GPT-2) in 2019.

4

External Validity

The benchmark's 170 tasks may not perfectly represent real-world intellectual labor distributions. Supplementary evidence suggests trends hold, but uncertainty remains.

5

Key Limitations

Poor performance on unstructured tasks; uncertain generalization to real-world job distributions. Both limit confidence in naïve forward extrapolations.